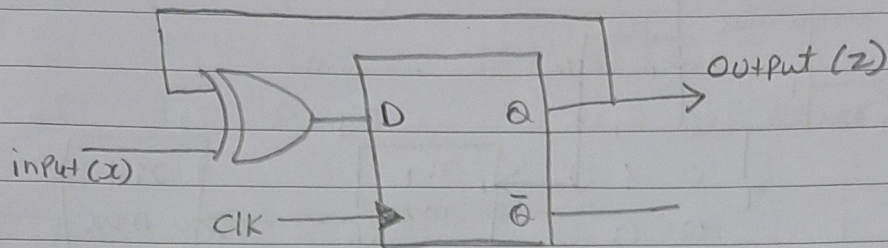


④ Sequential - Assignment.

1.) Explain the behaviour of the given circuit



From the above. $D = x \oplus Q$ and $Q_{next} = Q$

$\therefore$

Q	x	$D = x \oplus Q$	Q_{next}
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

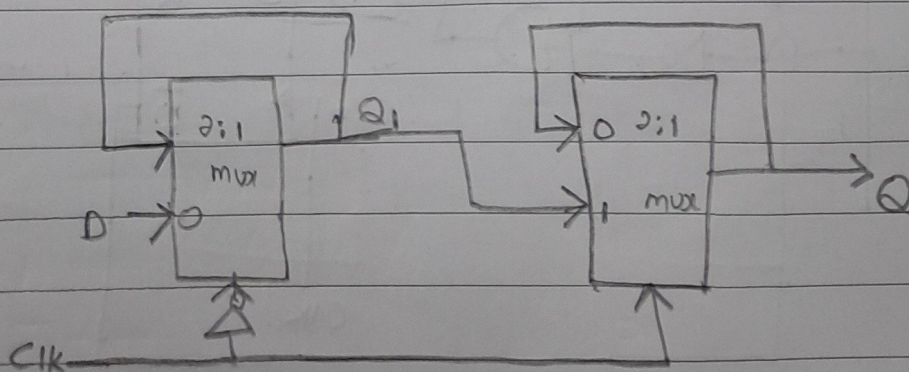
Comparing input x and the next state Q_{next} we can say that

when $x = 0$ $Q_{next} = Q$ and

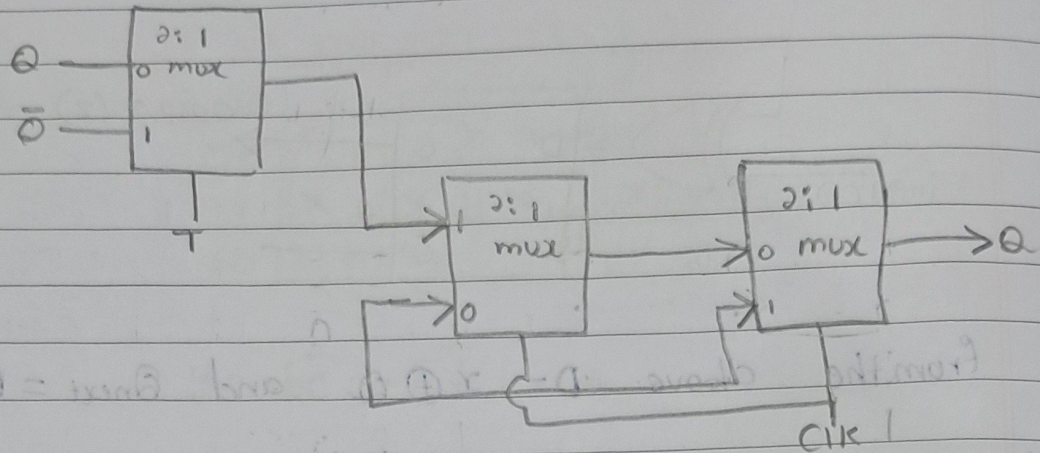
when $x = 1$ $Q_{next} = \bar{Q}$

$\therefore$ The behaviour of the given circuit is a T-flip-flop.

2.) Design a Posedge triggered D flip-flop using 2:1 mux only



3. Design a negedge triggered T flipflop using 2:1 Muxes only.



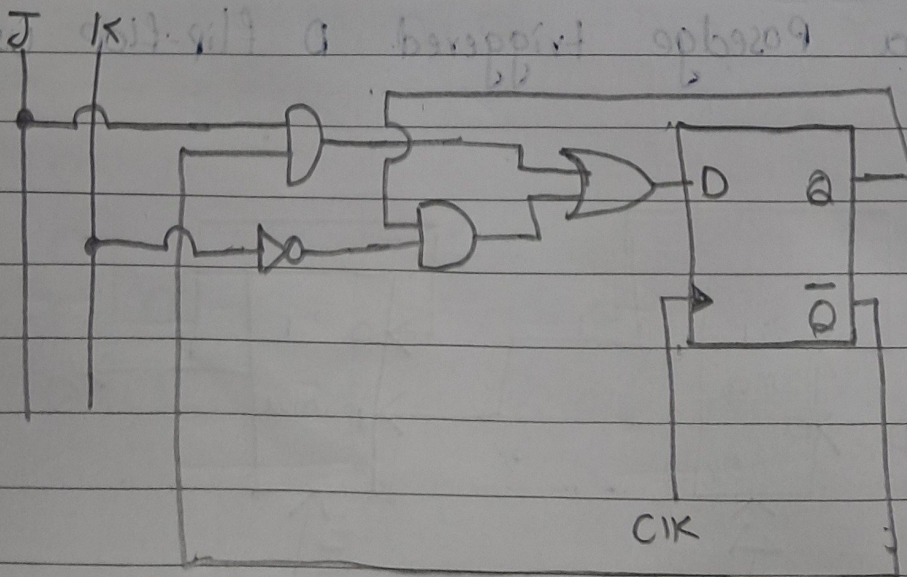
4. Convert by adding external gates:

a) A D flip-flop to a J-K flip-flop

J	K	Present state (Q)	Next state (Q _{next})
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

J	K	Q	Q _{next}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

$$D = Q_{next} = J\bar{Q} + \bar{K}Q$$



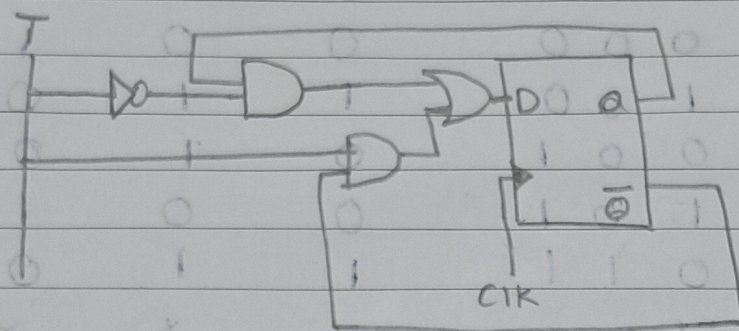
b) AT flip-flop to a D flip flop

T	Q	Q _{next}
0	0	0
0	1	1
1	0	1
1	1	0

T	Q	Q _{next}
0	0	0
0	1	1
1	0	1
1	1	0

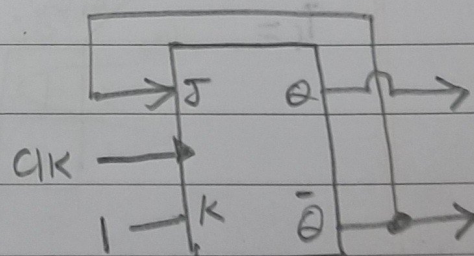
$$Q_{next} = \bar{T}Q + T\bar{Q}$$

or use xor gate.



5.) In a positive edge triggered JK flip-flop, we have $J = \bar{Q}$ and $K = 1$ as shown in the figure below. Assume the flip-flop was initially cleared and then clocked for 6 clock pulses. Find the sequence of the Q output for those 6 clock cycles.

Positive edge.	J	K	Q	$\bar{Q}$
1st	1	1	1	0
2nd	0	1	0	1
3rd	1	1	1	0
4th	0	1	0	1
5th	1	1	1	0
6th	0	1	0	1



∴ The sequence $Q = 1, 0, 1, 0, 1, 0$